

Rhyse Bendell, Ph.D.

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Human factors and cognitive systems researcher specializing in human–AI teaming, team cognition, and AI-supported decision making in complex environments. Designs and conducts instrumented human-subjects studies while also developing simulation testbeds, research software, and evaluation pipelines for emerging AI-enabled systems. Combines experimentation, computational modeling, and prototyping to investigate human–AI interaction and translate theory into practical tools and interventions.

EDUCATION

Ph.D. in Human Factors and Cognitive Psychology, **University of Central Florida**, FL, USA 01/21 - 12/23
M.A. in Applied Experimental & Human Factors Psychology, **University of Central Florida**, FL, USA 08/17 - 05/21
B.Sc. in Mechanical Engineering, **Massachusetts Institute of Technology**, MA, USA 08/12 - 05/16

RELEVANT EXPERIENCE

Postdoctoral Scholar, Cognitive Sciences Lab, University of Central Florida, FL, USA 01/24 - Current

- Evaluated LLM-enabled 'Course of Action GPT' in Army Research Laboratory with subject matter experts
- Designed and ran 2 subject matter expert-focused studies to assess interaction with generative AI tools
- Acted as co-project manager for AFOSR ~\$500,000 grant award, coordinating ~20 team members across 4 labs
- Produced workflow- and cognition-informed application design guidance for AI-supported sensemaking
- Solo-developed LLM-powered research assistant agent designed to support and automate research workflows
- Engineered human-in-the-loop, approval-gated AI workflows for literature review and synthesis
- Specified scalable evaluation models and experimental methods to identify workload drivers and failure modes
- Led research quantifying team cognition and processes to enable AI-driven performance interventions
- Co-developed analytic pipelines linking multimodal behavioral data to team cognitive models
- Ensured the timely completion of over 50+ DARPA, DOD, ARL, NSF, and AFOSR program deliverables
- Articulated and tested AI-enabled team knowledge-space management and measurement approaches
- Advised ~10 graduate research assistants, delegated tasking for data collection, processing, and lit. review

Graduate Research Assistant, Team Performance Lab, University of Central Florida, FL, USA 05/17 - 12/23

- Conducted 2 virtual reality experiments ($N = 274$) for dissertation with solo-built testbed
- Built 6 custom experimental VR testbeds in Unreal Engine to support controlled variable manipulation
- Selected by DARPA Program Manager to chair fast-track working group for experiment and testbed design
- Elected as head of the 40+ member interdisciplinary Experimentation Working Group for a DARPA program
- Established UCF's end-to-end analysis pipeline across 6 large-scale studies (max participants of $N = \sim 2000$)
- Identified and applied quantitative methods to assess real-time team coordination and adaptation
- Co-conceptualized profiling approaches that were significantly predictive of artificial social intelligence agent
- Collaborated in interdisciplinary profiling component agent integration with development team at MIT/DOLL
- Managed 2 human-subjects experiments assessing workload, interface usability, and trust in robotic teammates
- Facilitated technical and research deliverables across multiple labs, 10+ investigators and research assistants
- Developed high-fidelity VR simulation testbeds to evaluate multimodal human–robot interaction approaches
- Supported field-test evaluations examining gesture and voice control for robotic systems for army contexts
- Successfully met DARPA, DOD, and ARL funded research program deliverables for 3 projects

Contract Automated and Robotic Systems Integration Designer, Sanofi Pasteur, FL, USA 01/20 - 09/22

- Translated scientist requirements into integrated automated laboratory system designs
- Designed and simulated robotic enclosures, safety mechanisms, and interfaces using CAD (SolidWorks)
- Produced fabrication-ready drawings and part specifications for automated system hardware

Undergraduate Research Assistant, Bio-mechatronics Lab, MIT Media Lab, MA, USA 05/14 - 12/16

- Ran human-subjects biomechanics experiments using motion capture to evaluate assistive device performance
- Designed, fabricated and tested experimental mechatronic prosthetic prototypes with end users
- Wrote embedded software and control logic for prosthetic actuation, data collection, and evaluation

SKILLS AND PROFICIENCIES

Human Subjects & Experimentation: study design, IRB/protocol development, SME studies, multimodal data collection, behavioral/team-process coding

Analysis & Modeling: mixed-effects models, Bayesian methods, signal detection theory, mental-model analysis, motion/coordination analytics, agent-based/process modeling

AI, Simulation & Software: Python, Unreal Engine, C++, AI-supported workflow tools, research software, GUI development, automation pipelines, data visualization

Engineering & Instrumentation: SolidWorks, prototyping, embedded systems, hardware/software integration, eye tracking, motion capture, ECG, EEG, GSR

AWARDS

People's Choice Award, Four Minute Thesis Competition, University of Central Florida, FL, USA	11/25
UCF Preeminent Postdoctoral Program, University of Central Florida, FL, USA	01/25
Best Student Paper Award, Human Factors and Simulation track, AHFE 2023 Conference, CA, USA	07/23
Graduate Travel or Presentation Fellowships (x9), University of Central Florida, FL, USA	09/18 - 12/23
2 nd Place, Otronicon Game Jam, Orlando Science Center, FL, USA	01/19
Office of Research and Commercialization Doctoral Fellowship, University of Central Florida, FL, USA	09/17 - 09/21

PUBLICATIONS AND PRESENTATIONS

- *Publications:* 5 peer-reviewed journal articles (including 1 in Nature's Scientific Reports) and 3 book chapters
- *Conference Proceedings:* 25 peer-reviewed conference proceedings papers at international conferences
- *Invited Lectures:* 7 invited lectures to universities and organizations

SELECTED PUBLICATIONS [11/34]

1. **Bendell, R.**, Williams, J., Fiore, S. M., & Jentsch, F. (2025). Artificial social intelligence in teamwork: how team traits influence human-AI dynamics in complex tasks. *Frontiers in Robotics and AI*, 12, 1487883.
2. **Bendell, R.**, Williams, J., Fiore, S. M., & Jentsch, F. (2024). Individual and team profiling to support theory of mind in artificial social intelligence. *Scientific Reports*, 14(1), 12635.
3. **Bendell, R.** (2023). The Interplay of Spatial Ability, Sex, Training Modality, and Environmental Features: Effects on Spatial Cognition, Mental Map Formation, and Wayfinding. Graduate Thesis and Dissertation 2023-2024. 55. <https://stars.library.ucf.edu/etd2023/55>
4. **Bendell, R.**, & Williams, J. (2023). Assessing Spatial Knowledge and Mental Map Development Under Virtual Training Conditions. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 67(1), 1611-1616. <https://doi.org/10.1177/21695067231192219>
5. **Bendell, R.**, Williams, J., Fiore, S., Jentsch, F. (2023). Interventions by Artificial Socially Intelligent Agents in Collaborative Environments: Impacts on Team Performance and Knowledge Externalization. In: Waldemar Karwowski and Tareq Ahram (eds) *Artificial Intelligence, Social Computing and Wearable Technologies*. AHFE (2023) International Conference. AHFE Open Access, vol 113. AHFE International, USA. <http://doi.org/10.54941/ahfe1004189>
6. **Bendell, R.**, Williams, J., Fiore, S. M., & Jentsch, F. (2023). Teamwork Traits Associated with Positive Perceptions of the Dependability and Utility of Autonomous Advisors. Proceedings of the *Human Factors and Ergonomics Society Annual Meeting*, 67(1), 440-445. <https://doi.org/10.1177/21695067231192207>
7. Williams, J., **Bendell, R.**, Fiore, S., & Jentsch, F. (2023). The Role of Artificial Theory of Mind in Supporting Human-Agent Teaming Interactions. *Human Factors and Simulation*, 83(83).
8. **Bendell, R.**, Williams, J., Fiore, S., & Jentsch, F. (2023). Participant Gaming Experience Predicts Mental Model Formation, Task Performance, and Teaming Behavior in Simulated Search and Rescue. *Human Factors and Simulation*, 83(83).
9. Fiore, S.M., **Bendell, R.**, & Williams, J., (2023). Virtual Teams and Team Cognition. In L. Gilson, T. Maynard, & T. O'Neill (Eds.), *Handbook of Virtual Work* (pp. 280-304). Cheltenham, United Kingdom: Edward Elgar Publishing.
10. Howard, T., Stump, E., Fink, J., Arkin, J., Paul, R., Park, D., Roy, S., Barber, D., **Bendell, R.**, Schmeckpeper, K., Tian, J., Oh, J., Wigness, M., Quang, L., Rothrock, B., Nash, J., Walter, M., Jentsch, F., & Roy, N. (2022). An Intelligence Architecture for Grounded Language Communication with Field Robots. *Field Robotics*, 2, 468-512.
11. Maimon, B., Zorzos, A., **Bendell, R.**, Harding, A., Fahmi, M., Srinivasan, S., Calvaresi, P., Herr, H. (2017). Transdermal optogenetic peripheral nerve stimulation. *Journal of Neural Engineering*, 14(3)